SPES | Sustainability performances, evidence & scenarios

D8.3 SPES Co-Design Workshops: A Partner's Manual for Implementing Le Grand Jeu, Debriefing, and Follow-Up Interviews

January 2026

Funded by European Union's Horizon Europe Programme under Grant Agreement No. 101044511

Authors

Simen Mørstad Johansen–Researcher of the SPES Project, Oslo Metropolitan University

Federico Bonelli – Researcher of the SPES Project, University of Amsterdam

Toa Giroletti – Researcher of the SPES Project, University of Florence

Andrea Ferrannini – Researcher of the SPES Project, University of Florence

Mi Ah Schoyen – Team leader and Researcher of the SPES Project, Oslo Metropolitan University

Contributors and peer reviewers

Stefania Milan – National team leader of the SPES Project, Universty of Amsterdam**; Vaqar Ahmed** – Researcher of the SPES project, PEPdelete-Partnership for Economic Policy**; Anikó Bernát** – Researcher of the SPES Project, TÁRKI Institute for Social Research**; Menaca Calyaneratne – Researcher of the SPES project, PEP-Partnership for Economic Policy; Filippo Cuccaro** – Researcher of the SPES Project, University of Florence**; Nilakshi De Silva** – Researcher of the SPES project, PEP-Partnership for Economic Policy**; Neranjana Gunatilleke** – Researcher of the SPES project, PEP-Partnership for Economic Policy**; Meg Kitamura** – Researcher of the SPES Project, University of Amsterdam**; Daniyal Masood** – Researcher of the SPES project, PEP-Partnership for Economic Policy**; Marion Ouma** – Researcher of the SPES project, PEP-Partnership for Economic Policy

Acknowledgements

This manual is part of a broader methodology developed for participatory workshops under the SPES project. These workshops aim to explore sustainable human development through innovative tools such as Le Grand Jeu (LGJ), focus group discussions, and follow-up interviews. By combining immersive gameplay with structured group discussions and individual reflections, the SPES workshops facilitate a comprehensive understanding of aspirations, challenges, and policy solutions in the transition to sustainability.

The creation and evolution of LGJ owes its success to a collaborative effort spanning years and numerous contributions. We extend our heartfelt gratitude to the dedicated co-design teams, test players, and workshop participants who generously shared their insights, creativity, and critical feedback to refine the game and its scenarios.

The integration of LGJ into participatory workshops was enriched by co-design sessions in Florence, Budapest, and online. Thanks to all the contributors whose perspectives and contributions shaped not only the adaptation of LGJ to the purposes of SPES but also the focus group and interview methodologies for addressing the challenges of Sustainable Human Development. Thanks also to colleagues and friends who participated in additional pilot and training sessions.

Lastly, we acknowledge the pioneering work of Federico Bonelli and Raffaella Rovida (Petz et al., 2024), whose vision brought LGJ to life.

Cite as

Johansen, SM., Bonelli, F., Giroletti, T., Ferrannini, A., Schoyen, M.A. (2026). *SPES Co-Design Workshops: A Partner's Manual for Implementing Le Grand Jeu, Debriefing, and Follow-Up Interviews*. SPES project – Sustainability Performances, Evidence and Scenarios. Florence: University of Florence. Available at: https://www.sustainabilityperformances.eu/publications-deliverables/

Disclaimer

This working paper for the project SPES has been prepared by Oslo Metropolitan University, the University of Amsterdam, and the University of Florence for Work Package 8 related to Task 8.3. This methodological manual has enabled SPES project partners to design and implement participatory workshops that explore sustainability transitions and Sustainable Human Development, by combining immersive Le Grand Jeu game sessions, structured group discussions, and individual reflections conducted throughout the SPES project activities.

This deliverable contains original unpublished work except where clearly indicated otherwise. Acknowledgement of previously published material and of the work of others has been made through appropriate citation, quotation or both. Generative AI was used in the work with this manual. The information and views set out in this report are those of the authors and do not necessarily reflect the official opinion of the Commission. Neither the Commission nor any person acting on the Commission's behalf may be held responsible for the use which may be made of the information contained therein.

The project SPES is funded by the European Union's Horizon Europe Programme under Grant Agreement No. 101094551.

Table of Contents

Manual description 6
Introduction to the workshops 7
Research Questions Addressed in the Workshops 7
Outputs and outcomes from the Workshops 8
How to use this manual 8
Section 1: Arranging the workshops: playing LGJ, focus groups and follow-up interviews 9
1. Before having the workshop 9
1.1. Recruitment of participants 9
1.2. Ethical approval and data management 10
1.3. Information to participants and Consent Forms 10
2. Before playing the game 11
2.1. Making the physical game 11
2.2. Research team: The game master and the observer 11
2.1.1. The Game Master's Role: facilitating the game 11
2.1.2. The Observer's role: documentation of the game 11
2.3. Guidance for the masters and observers "training the trainers" 12
3.Playing the game 13
3.1. The game kit 13
3.1.1. Board 13
3.1.2. Materials 14
3.2. Script to introduce the Game to the players 15
3.3. Playing the game 16
3.3.1. Distribute Starting Resources: 16
3.3.2. First Round 17
3.3.3. Subsequent Round (From Round 2 Onwards) 17
3.3.4. End of Game 19
3.3.5. Gameplay Tips 19

3. After playing the game: focused group discussions 20
4.1. Aspirations, Problems and Policy Recommendations 20
4.2. Template for focus discussions 21
4.3. Documentation of the focused discussion 22
4.4. Field Notes from the Workshop 23
5. Post-focus groups individual debrief/interview 23
5.1. Guide for Semi-Structured Follow-Up Interview 23
5.2. Documentation of the individual interviews 25
Section 2. After the workshop: Reporting 26
6. Writing the national report 26
6.1. Template for the national workshop reports 26
Section 3. 28
7. Appendices 28
7.1. Events 28
7.1.1. Collective event cards (LGJ) 28
7.1.2. Individual cards 30
7.2. How to use the dice 32
7.2. How to use the wheel 34
7.3. Information letter to participants 35
How can productivity, equity, environmental sustainability, participation and empowerment, and human security come together to drive the transition towards sustainable human development in Europe? 35
8. References 43

Manual description

This manual details the methodology developed and applied within the SPES project for participatory workshops. It provides step-by-step guidance for researchers and practitioners on implementing immersive Le Grand Jeu (LGJ) game sessions, facilitating structured post-game group discussions, and conducting optional follow-up interviews. By combining interactive gameplay with reflective discussions and individual reflections, this approach enables participants to explore complex societal dynamics and co-create insights into the challenges and opportunities of sustainable human development.

Designed based on the practical experience of SPES project partners, the manual captures the full process as it was applied in real workshops, from participant recruitment and game facilitation to debriefing and data collection. It supports researchers in transforming participants' experiences and discussions into actionable knowledge, highlighting their aspirations, perceived barriers, and potential policy solutions. By integrating immersive play with collaborative reflection, this methodology fosters a deeper understanding of sustainability transitions and provides a tested framework for engaging diverse stakeholders in co-design processes.

Links

- **SPES Project Website**: https://www.sustainabilityperformances.eu/
- **Le Grand Jeu Website**: www.legrandjeu.net and GitHub.

Licensing

- © LGJ 2016-2024: Federico Bonelli – Raffaella Rovida.
- LGJ is licensed under **Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License**.
- The SPES LGJ scenario is co-designed and licensed to the SPES consortium under the same license, **free to use for non-commercial purposes worldwide**.

Introduction to the workshops

This manual is designed to guide researchers in planning, executing, and analysing workshops centred on *Le Grand Jeu (LGJ)* and the following debriefing activities. These workshops are integral to the SPES project (Sustainability Performance, Evidence, and Scenarios), which seeks to understand how to reconcile productivity, equity, environmental sustainability, participation, and human security in the transition towards sustainable human development in Europe. Thus, at the heart of the SPES context lies the complex challenge of balancing the economic, social, and ecological dimensions of sustainability.

The workshops consist of three main activities:

1. **Playing Le Grand Jeu (LGJ)**: A participatory and narrative-driven serious game that simulates societal dynamics, allowing participants to explore challenges and co-create solutions for transitions to sustainable societies.
2. **Focused Discussions:** Facilitated group interviews where participants reflect on the game, bridging their in-game experiences with real-world aspirations, challenges, and policy recommendations.
3. **Follow-up individual interviews (optional)**

These workshops create a structured yet creative space for participants to grapple with trade-offs, synergies, and tensions across these dimensions, fostering deeper understanding and collaborative problem-solving.

Research Questions Addressed in the Workshops

These activities aim to address key research questions central to work package 8 of the SPES project:

- **Aspirations**: What do participants envision as the values and goals of a sustainable society?
- **Problems**: What are the perceived barriers to achieving sustainability and equity in participants' local and national contexts?
- **Policy Solutions**: What actions, policies, or strategies do participants identify as critical for overcoming these challenges?

The integration of gameplay and reflection provides a unique framework for participants to articulate diverse perspectives, grounded in their lived experiences, and generate actionable insights.

Outputs and outcomes from the Workshops

The workshops are not just participatory exercises; they contribute directly to the research outputs of the SPES project. The objective of these workshop activities is primarily to produce data that can shed light on the aspirations and problems with climate transition policy making in the different national contexts, and recommend policy solutions.

The data from the workshops are produced through both "field observations" of the game session itself and the debriefing of the game, which consists of group and individual interviews.

Debriefing is a critical component of serious games, transforming gameplay experiences into meaningful insights. It serves as the bridge between the simulated world of the game and real-world contexts, allowing participants to reflect on their actions, decisions, and interactions. Effective debriefing involves multiple levels of reflection, from analysing in-game activities to interpreting their broader societal relevance.

Researchers will synthesize the thick data from the workshops (from the game, the group interviews/focused discussions, and individual follow-up interviews) to produce:

1. **National Reports**: Each country team will develop a concise report summarizing key themes, including aspirations, problems, and policy recommendations emerging from the workshops. These reports form the basis for comparative analysis across the participating countries.
2. **Working Paper**: The national reports will feed into Deliverable 8.4, a working paper tentatively titled *"The Transition from Below: Collectively Agreed Perspectives on Aspirations, Problems, and Policy Solutions."* This paper will provide a cross-national synthesis of the findings, highlighting shared challenges and localized strategies.

By engaging diverse participants and focusing on co-creation, these workshops provide rich qualitative data and foster dialogue on the societal transformations needed for sustainable human development. This manual offers step-by-step guidance to ensure researchers can effectively implement and document this process.

Finally, a secondary objective also concerns the method of serious games workshops in itself, that is the use of serious games to create a shared understanding of climate transition challenges and co-design of policy solutions.

How to use this manual

This research manual has three main sections:

Section 1 (Chapter 1 - Chapter 5) concerns a step-by-step implementation of the method from recruitment of participants, introduction to the topic, how to play the game, and the debriefing session.

Section 2 (Chapter 6) concerns what the researchers do after completing a workshop concerning the transcription of the data, analysis and reporting.

Section 3 (Chapter 7 - Chapter 8) contains references and appendices with event cards.

Section 1: Arranging the workshops: playing LGJ, focus groups and follow-up interviews

The workshops in the SPES project WP 8 consist of three parts: First, playing Le Grand Jeu (LGJ). Second, having approximately 1 hour of focused group discussions immediately after playing the game. Third, short individual follow-up interviews 3-5 days after the workshop

This section of the research manual addresses how to arrange the workshops, that is, how to play the game, how to facilitate the focus group discussions and how to approach the short individual interviews.

1. Before having the workshop

1.1. Recruitment of participants

You should recruit 4-6 participants for each workshop. The recruitment aims to reflect different interests and vulnerabilities in relation to climate transition policies. Hence, you should try to achieve some degree of socio-economic, occupational and cultural diversity among the participants in the workshops.

To ensure meaningful engagement, we take a pragmatic and context-sensitive approach to participant recruitment that embraces the political and cultural differences across countries. Each country team determines its own criteria and strategy for selecting participants, tailoring the process to local contexts and considerations. This flexibility allows us to adapt the game sessions to the unique dynamics of each region while ensuring relevant and diverse perspectives are represented.

For instance, in Norway and the Netherlands, the recruitment strategy focuses on two distinct groups: one comprising NGOs representing diverse population interests in the green transition, and the other of “ordinary” citizens, recruited through public systems such as libraries (e.g., in Amsterdam). By contrast, in Hungary, the TÁRKI team anticipates that they can easily engage citizens with low socioeconomic status but find it less relevant to include representatives from civil society organizations. These country-specific approaches reflect our commitment to inclusivity and contextual sensitivity in participant recruitment.

1.2. Ethical approval and data management

Each team is responsible for treating personal data in accordance with the EU General Data Protection Regulation (or corresponding national legislation in the case of non-European countries), as well as with EU, national and (when applicable) local standards for research ethics.

1.3. Information to participants and Consent Forms

A short email or one-page document is shared with all participants to set common parameters on the main elements of the game

This email contains:

- Information about SPES and the aim of the research.
    - The written information on SPES can be supplemented by a short film in the local language about climate change and transition to set the tone. As an example, this link goes to the film we will use in Norway. National teams have to find appropriate films for their context.
- Information about the focus group and LGJ,
    - The scope of the whole session
    - The basics of and intention of playing the game, including a link to a video on the basics of LGJ (https://vimeo.com/477278036).
    - About the collection of data, pictures, sound and notes.
- A consent form that can be signed and returned, or signed upon arrival.
    - Consent has to be given explicitly to process personal data (including audio and/or video recordings). This should be handled according to relevant European, national and (if applicable) local legal requirements.

An example of such an information letter and consent form to the participants is included in the appendix section 8.4. This letter can be adapted and translated into the local language and context by the researchers.

2. Before playing the game

2.1. Making the physical game

See files in the appendix, and git hub here

2.2. Research team: The game master and the observer

The research team consists of a minimum of two people. During the playing of the game at the workshop, the two people fill the roles of game master and observer. During the following focus group discussions of the workshop, these two people facilitated.

2.1.1. The Game Master's Role: facilitating the game

Game Master (GM): Serves as the facilitator and narrator of the game, guiding players through scenarios, resolving events, and ensuring the rules are applied fairly. The GM plays a pivotal role in shaping the narrative and maintaining the flow of the game.

The GM is central to the success of LGJ, acting as a facilitator, modelling the ideas and proposals from the players into the game language. The GM ensures that gameplay remains engaging and immersive. The GM sets the tone, introducing the SPES scenario and encouraging players to vividly describe their actions to co-create an engaging narrative. The GM manages gameplay by overseeing resources, enforcing players' decisions, and using tools like dice, event cards, and the wheel to simulate real-world challenges.

More detailed information is in Chapter 4.

2.1.2. The Observer's role: documentation of the game

The observer is responsible for documenting how the game evolves. The observer will:

1. Take the role of prompter/assistant for the GM during the game to help keep track of what has happened and decisions that have been taken in the game.
2. The observer's documentation of the game will serve as the foundation for the focus group discussion following the game.
3. The documentation of the game will be a part of the material to be analysed for the national reports.

The game is documented on the go by the observer. The role of the observer is similar to that of a field researcher. Documenting the game entails taking detailed notes. The observer can take pictures of actions and outcomes on the board (avoiding participants' faces) and use post-it and colored markers to document players' agreements, policy decisions, problem-solving approaches, and identify attitudes and roles, as well as other relevant points. After the workshop, the observer should produce "field notes" from the game. See section 4.3 for more information.

This documentation of the game will be a part of the wider material that will serve as the basis for the analysis in your country report, supplementing the documentation of the focused discussion and the follow-up individual interviews.

In addition, you may audio record and/or film the game sessions and use them as a tool to improve your written notes and give you richer material to analyse after the game. However, if you do so, it is important to remember that videos and audio recordings of individuals qualify as personal data and, hence, under the European data protection legislation (GDPR art. 6), you need a legal basis for processing such data. In the non-European countries, you need to ensure that you follow the relevant national (and if applicable local) legislation on data protection.

2.3. Guidance for the masters and observers “training the trainers”

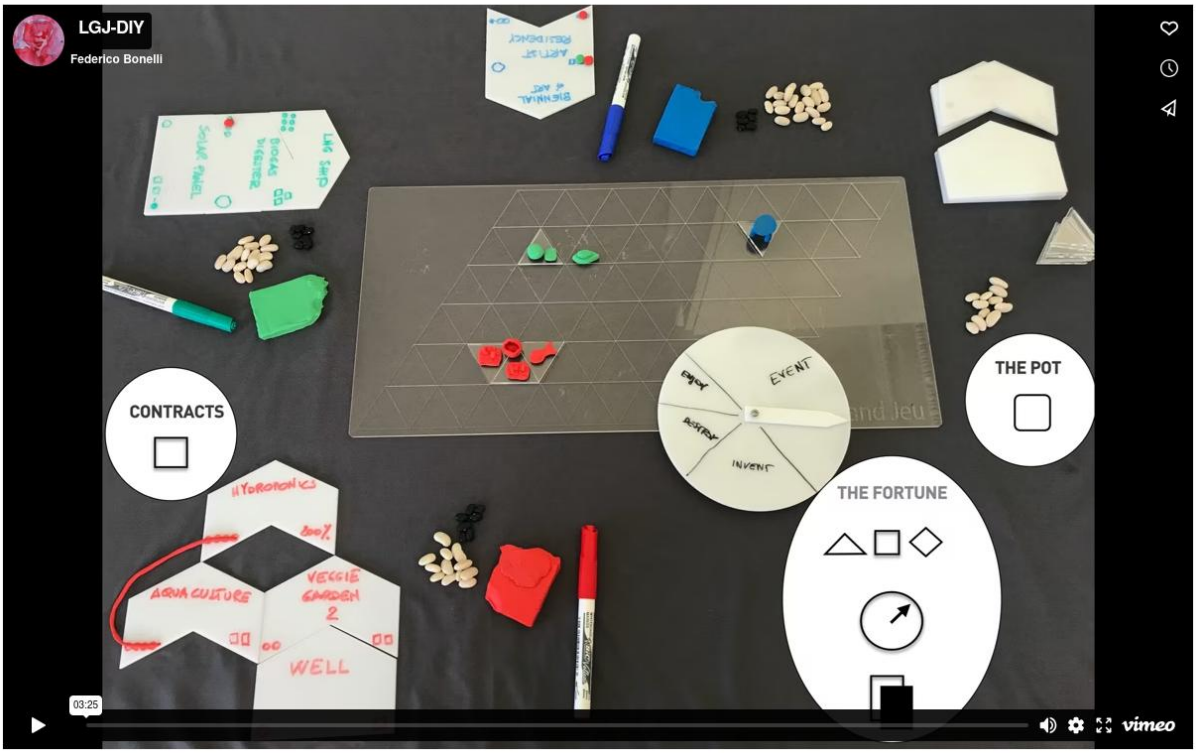
We provide training to help researchers take on the roles of observer and game master for the SPES scenario on LGJ. This includes the LGJ manual, literature, game examples (from the LGJ website), remote access to game scenario tests, and in-person instruction.

3.Playing the game

3.1. The game kit

This section gives an overview of the physical parts of LGJ. How these parts are applied to play the game is described and explained in the subsequent sections. For a quick introduction to the game kit and the game, see https://vimeo.com/477278036

DO IT YOURSELF



3.1.1. Board

The Board consists of similar pieces of diamond shaped grid of equilateral triangles. These can be made of transparent plastic or of paper on cardboard.

The Grid is the mesh of triangles drawn on the board. On the grid, you have (subject to tax payments) access to water, electricity, roads and the internet, as well as the other basic social services such as hospitals, schools that we deem part of civilization. The GM can put some symbols on the board to show the presence of public institutions.

3.1.2. Materials

Land: transparent triangles are to be used to represent the spaces that you control in the game. Play-dough is used to mark ownership of the modelled activity. If you set your triangle(s) on the grid means that you are connected to the societal infrastructure; if you set your triangle(s) off the grid, you are not connected.

Elements/Contracts: The elements are used to keep a record of the economic activities of the players. The cost of white and grey is noted on the elements.

Play-dough: Players are given play-dough of different colours to mark their property (land with their "projects").

Tokens: Physical markers for whites, greys (e.g., beans, stones, or tokens), and social value. The colours are suggestions and can be decided upon locally.

- White: currency, resources, knowledge, energy, etc.
- Grey: pollution, loss of biodiversity, depletion of resources, entropy, CO2 accounting, etc.

Social value scale: A scale from -10 to +10 that is drawn on paper. This is used to measure the social value of the fictional society. Social value could refer to the themes under the 5 pillars of sustainable society. A token is used to indicate where on the scale the society is.

-10					-5					0					5					+10

Transparent bowls:

- For the collective local amount of white and greys.
- For the individual amounts of white and grey.

Bag of white and grey tokens: This is the grand pot

Dice: A variety of dice (d4, d6, d10, etc.) for resolving player actions and events. d4 means a dice with four sides, d6 with six sides, etc. See Appendix 8.2 for more information

Event Cards: Written cards that introduce unexpected events to the game. There are:

- collective event cards: these affect all players
- individual event cards: these affect only the individual player.

The events and their consequences are discussed with the players and modelled with their consequences on the game by the master. See Appendix 8.1 for more information

The Wheel: A spinner to introduce chance and trigger events. Starts by 80% chance of a collective event; 20% chance of an individual event. There are also further possible uses of the wheel to model complex events, see Appendix 8.3.

Hourglass: To keep track of time during discussions (optional).

Markers: To write on the board and the elements

Pen and paper: Players are given a notebook and a pen to help them keep track of their own and the overall game, as well as ideas and reflections.

3.2. Script to introduce the Game to the players

The basic SPES setting to play the game is described as follows.

Each of you has recently decided to move to live in a mid-sized city composed of two main areas:

A) A dynamic core area of the city where several business activities, the main financial services, the city hall and government building, the main public school and hospital are located [anything else can be added by the GM here]. Living standards are higher here, but there are inequality issues within the population.

B) A rural area characterized mainly by green/restorable lands, with limited services. Living standards are somewhat lower here, but the community is more cohesive due to less inequality.

These two areas are separated by a navigable river/canal/lake [to be adapted by GM to the local context] and surrounded by mountains / wilder areas [to be adapted by the GM to the local context].

Both areas are populated by overall friendly people and offer some basic grid facilities/infrastructure (e.g., internet, water, electricity) and basic social services. Such facilities and services are not available if you decide to live in the wilder areas that are off-the-grid. The urban core and rural area are separated by a navigable river or canal, with mountainous wilderness areas surrounding the city. Players can choose to establish their presence in any of these zones, each with distinct benefits and trade-offs. Players can set their land on or off the grid. If players choose a wilder area off the grid, they have no access to utility, and they pay no taxes. The master adjusts their game to this fact. The players have to live in this fictional world and make decisions about their lives. The player with the most whites when subtracting the greys wins.

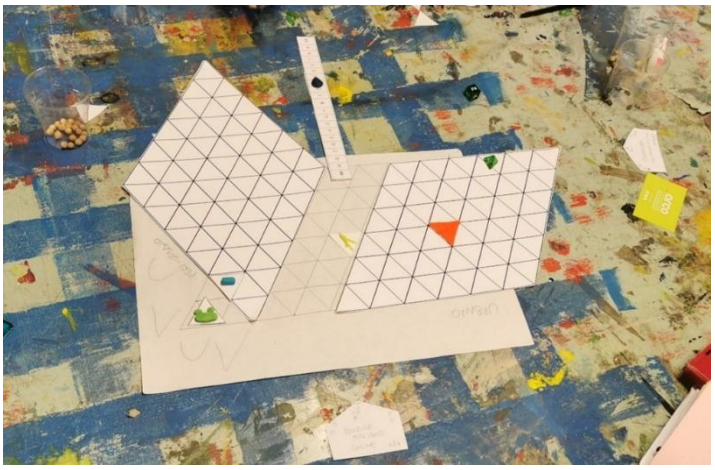


Image 1: *Overview of SPES setup*

The resources and consequences of individual actions are operationalized in white and grey tokens. There are three levels of these:

- **The individual pots of tokens:**
    - White - Players can gain white. These are placed in their individual bowl for white and can be used by the player.
    - Greys - players can receive greys. These are placed in their individual bowls/glass for grey. These are aggregated in the bowl/glass throughout the game.
- **The collective pots of tokens**:
    - White: These are the aggregated taxes that can be used by the collectivity of players upon calling an assembly

  - Grey. Greys that are related to local pollution of the player are allocated in the individual pot; those greys which are allocated as collective pollution/emission are allocated to the collective pot. The distribution is decided by the GM.
  - Social value ladder: This is an operationalization of the collective assessed social atmosphere in the created society. A "ladder" is drawn up on a paper from -10 to + 10. A token is used to indicate the current social value. Starts at 0 and is adjusted for each round.

- **The Grand pot of Le Jeu**: This is the storage of whites and greys that is controlled by the GM. This could be understood as the "market" and is not accessible for collective actions.

End game

1) If a player has no white tokens, that player is out of the game.
2) If greys in the collective pot are more than the number of players multiplied by 10, the game ends (because of the disastrous ecological conditions); for example, if there are 4 players, the upper limit is 40 greys.
3) If social value is at -10, the game ends (because of the disastrous social conditions).

The players can create their own roles that fit in a broad sense within the description of the fictional world of the LGJ SPES scenario.

3.3. Playing the game

Each gameplay round consists of actions taken by players, followed by collective evaluations and event resolutions. Rounds are designed to simulate real-world decision-making and resource management in the SPES scenario.

3.3.1. Distribute Starting Resources:

- **Optional:** The GM can choose to start with an event card to further specify the scenario (and get quicker into the relevant discussions)
- **Land**: Each player rolls a 1d4 +1 to determine the number of land units they control at the start. Players can buy or trade additional land during the game.
- **Individual pots.** Players allocate these resources in their personal pots, representing their initial assets and liabilities.
  - **Whites**: Roll 2d6 to determine the starting number of whites.
  - **Greys**: Roll 1d6 to determine the starting number of greys.

3.3.2. First Round

Introducing character, setting up land and activity:

- Each player introduces their character, outlining their motivations, initial goals, and chosen starting location on the map.
- Players place their initial land units on the game board, selecting locations on-grid (urban/rural areas) or off-grid (wilderness).
- Ask the players to put on their land a piece of play-dough with their colour, to recognize them, and then to mimic the thing they will build, adding and moulding more details as the game progresses.

Estimating and paying initial Costs. Based on the description of the activity of the player, the GM estimates the initial Cost (W), initial taxes (W), and initiation externalities (G) that are done only at the set-up of land.

1. Pay taxes to the collective pot according to the area
    a. Dynamic Core area: 2
    b. Rural area: 1
    c. Off the grid: 0
2. receive externality from the Grand Pot
    a. Dynamic Core Area: 1
    b. Rural Area: 2
    c. off the grid: 0
3. Define and pay the costs to the grand pot to build your activity
    a. This is defined by the GM.
4. Define the costs to pay to maintain your activity to be paid at each round to the grand pot
        i. This is also defined by the GM
5. Throw a dice 1d10 to estimate your potential for revenues. Based on the percentage on the dice and the description of the activity, the GM decides what dice to use in future rounds (d4; d6; d8; d10; d12; d20). If high probability of success (over 50%), the dices with large range (that is d10; d12; and d20) lead to potentially higher revenue. If low probability of success, the dice with low range (d4; d6; and d8) can be used. See Appendix 8.2.

3.3.3. Subsequent Round (From Round 2 Onwards)

During their turn, players can perform one of the following actions:

Invest:

- Expand operations by adding new land or upgrading existing activities.
- The Player rolls a d10 to determine the probability of success of his endeavour, the GM adjusting the story to the result of the dice throw, models costs (whites), revenues (whites), and externalities (greys). See Appendix 8.2.

Invent:

- Propose and develop innovative solutions to economic, societal or environmental challenges.
- The GM and players assess the plausibility and impact of the proposal and its cost in both energy and impact. Then allows a throw for the d10 to assess the success level of the action and models it into the game.

Call an Assembly:

- Propose collective actions or policies using the local pot of whites (e.g., infrastructure investments, welfare measures).
- Players debate and vote on the proposal, guided by the GM. Decisions may generate greys or require redistribution of whites.

React to Other Players:

- Engage in trade, alliances, or other interpersonal interactions to advance goals collaboratively or competitively.

End-of-Round Actions

Event Resolution:

- One player spins the wheel to determine whether a collective or individual event occurs. Use a dice to determine which player spins the wheel
- Draw a card from the deck of cards or roll a dice to select the event. Read the card for the player(s) and initiate discussions accordingly before subsequently resolving the consequences, as operationalized in the game tokens, for the affected players or community. See Appendix 10.1. for detailed information on how to proceed.

Social Value Assessment:

- Players collectively evaluate the round's impact on social value (scale: -2 to +2).
- Based on this evaluation, add the quantified social value created or destroyed to the local social value scale. The social value in the community is kept note of by the GM on the social value scale.

observer **4.4. Ongoing Resource Management**

At the start of each new round:

- Players pay recurring costs (whites for maintenance, taxes) for their established activities.
- Revenues and externalities from ongoing actions are calculated and distributed

3.3.4. End of Game

The game concludes after 2 hours, when usually four-five rounds have been completed. The game also stops if an ecological collapse occurs due to the collective greys exceeding the ecological ceiling.

At the end, the Game Master facilitates a debrief session to reflect on the outcome, see Chapter 8.

We suggest allowing a 15-minute break after the game for coffee/toilet, and fruit before starting the debriefing session.

3.3.5. Gameplay Tips

To ensure a smooth and engaging game, the GM should consider the following tips:

1. **Foster Immersion**:
    a. Encourage players to describe their actions in detail, including motivations and impacts.
    b. Use descriptive language to make the game world feel vivid and real.
2. **Balance Game Pace**:
    a. Keep rounds manageable by gently steering players back on track if discussions become too long or tangential.
    b. Use a timer or visual cue to limit decision-making time when necessary.
3. **Facilitate Collaboration and Debate**:
    a. During assemblies, guide discussions by posing questions and summarizing points of agreement.
    b. Ensure that quieter players have opportunities to contribute.
4. **Leverage Game Tools Effectively**:
    a. Use the dice and wheel to add excitement and simulate chance events.
    b. Introduce event cards thoughtfully, tailoring their use to the narrative flow of the game.
5. **Adapt to the Players**:
    - Be prepared to adjust rules or elements to suit the group's dynamics or preferences.
    - Maintain flexibility, prioritizing fun and engagement.

3. After playing the game: focused group discussions

The debriefing sessions can be characterized as a group interview or a focus group. This is a focused discussion between the participants on the SPES thematic.

The roles of the researcher for the group discussion:

Facilitator/moderator: Guides the discussion, ensures all voices are heard, asks probing questions, and maintains focus on key topics.

Observer/assistant facilitator: Watches the group dynamics, taking detailed field notes on interactions, body language, and group decision-making.

4.1. Aspirations, Problems and Policy Recommendations

First, focus on the participants' experience of playing LGJ, and how their experiences can serve as a bridge to discuss real-life challenges with crafting sustainable societies.

Players review their individual achievements in light of their original aspirations and the potential reasons that hindered or helped them to reach their goals, and the overall societal performance, discussing the balance (or imbalance) between sustainability, equity, and productivity achieved during the game. This moment provides closure, allowing participants to connect their in-game decisions with broader themes of societal development.

These reflections set the stage for the subsequent focus group discussion, where participants can delve deeper into the aspirations, challenges, and policy solutions explored during the game.

In 4.1 is a proposed **template for the semi-structured group discussion** based on the game's context and experience, transitioning into aspirations, problems, and policy solutions in participants' national contexts. The template should be translated and adapted to your local language. In the template, there are six main sections: introduction, discussion of the central themes and outroduction. There is a distinction between core prompts to get the group to focus on the specific theme and optional prompts to dig deeper into the theme. Remember that the most important thing is to touch upon the three different themes, and that template mainly serves as a tool to do that. If the discussion naturally touches upon aspirations, problems and solutions, there is no need to intervene with these questions.

4.2. Template for focus discussions

1. Introduction and Bridge from the Game

Appreciation for participating in the game

As the first step in the focus group discussion, the game observer should provide a concise summary of the game session to help participants transition from gameplay to reflective discussion. Highlight the key moments, decisions, and challenges that emerged during the game, emphasizing the dynamics between individual strategies and collective outcomes. Thereafter, start involving the participants.

- **Proposed prompt**: "Let's start by reflecting on your experience during the game. What stood out for you as key challenges in creating a sustainable society in the game?"

Transition:

- **Proposed prompt**: "Now, let's take what we explored in the game and connect it to real-world situations of climate transition policies in [your country, e.g., Norway]. Imagine your country as one of the lands you managed—what similarities or differences do you notice?"

2. Aspirations for a Sustainable Society

Objective: Identify participants' vision for their society's future.

Core Questions:

- Imagine a truly sustainable and thriving society in your national context. How would it function in terms of energy production, economic equity, environmental protection, and social well-being? What specific policies, technologies, or cultural shifts would be necessary to achieve this vision?"

Optional additional prompts:

  - "What values or priorities should guide your society's transition to sustainability?"
  - "What lessons or ideas from the game would you like to see applied in your own life, in your local community or at national or international level?"

3. Identifying Key Problems

Objective: Explore barriers and challenges based on participants' lived experiences.

Core Questions:

- "In your view, what are the biggest barriers to achieving a sustainable society in your country?"
- "What groups or regions in your society are most affected by these challenges?"

Additional optional prompts:

  - "During the game, were there moments where certain decisions felt impossible or particularly challenging? How do those compare to real-life challenges you face?"

4. Exploring Policy Solutions

Objective: Co-create ideas for actionable change.

Core Questions:

- "Based on what we've discussed, what policies or strategies do you think could address these challenges?" "What role do you see for civil society, government, or private sectors and academia in implementing these solutions?"

Additional optional prompts

  - "During the game, were there any strategies that worked particularly well that could inspire real-world solutions?"

5. Reflections on Participation and Co-Design

Objective: Evaluate inclusivity and identify enablers of effective participation.

Core Questions:

- "How can we ensure that diverse voices, including those from socio-economically disadvantaged groups, are included in these solutions?"

Additional optional prompt

  - "What support would you need to participate in or advocate for these changes?"

6. Closing and Next Steps

Thank participants for their input and time.

Outline how their insights will be used.

Encourage ongoing reflection and involvement:

- **Prompt**: "What is one takeaway or action point you feel inspired to pursue after this discussion?"

Additional Notes for Facilitation

- Encourage participants to share specific examples and lived experiences.
- Use visual aids or references from the game to help bridge abstract ideas to practical realities.
- Ensure that all participants are heard and given space to contribute
- Be prepared with follow-up questions to probe deeper into responses or clarify insights.

4.3. Documentation of the focused discussion

The focus-group discussions are recorded in audio, optionally on video. These recordings are transcribed. In addition, the researcher who is not facilitating the focus group discussion should make research notes during and after the session, supplementing the transcribed discussions.

The transcriptions are done in the local language, while short summaries can be provided in English.

4.4. Field Notes from the Workshop

In addition to the audio/video recording of the workshops, the researchers can take field notes to capture observations and reflections that emerge during the game session and group discussions. Field notes help researchers capture nuances that may not be evident in interviews or recordings, enriching the depth of qualitative analysis. They can be taken during and immediately after the workshops so that the reflections are fresh.

The field notes reflections should focus on:

- The substantial SPES thematic.
- The method (using games as a part of focus group methodology).

These field notes do not go directly into the written report but will help provide the account of the workshop and catalyze the analytical process.

Field notes containing sensitive data should be stored alongside the other sensitive data produced in the workshops.

5. Post-focus groups individual debrief/interview

After the game session and focus group, the participants will digest and reflect upon their experiences. We would like to capture their thoughts. These follow-up interviews are optional for the participants. After the focus group, the researchers are responsible for asking the participants if they are interested in this and finding the time to contact the participants that opt-in (approx. 3-5 days after)

Purpose: This interview aims to capture participants' reflections and insights following their experience with *LGJ* on the one hand and focusing on aspirations, perceived problems, and potential policies related to sustainability and equity on the other hand. This is a guide and does not have to be followed strictly. The most important thing is to allow the participants to share their thoughts and reflections openly.

5.1. Guide for Semi-Structured Follow-Up Interview

Opening

Welcome and Introduction (2-3 minutes):

- Thank the participants for their time and participation in the game session.

- Briefly outline the purpose of the interview: to gather their thoughts and reflections a few days after the game experience.
- Reassure them about confidentiality and that there are no right or wrong answers.

Warm-Up Question (1-2 minutes):

- "Since the session, have you had any new thoughts about the game or the issues it explored?"

Core Questions

Reflections on Aspirations (5 minutes):

- "What, if any, new ideas or aspirations came to mind after the game about how a sustainable and equitable society might look?"
- "Did the game influence how you think about your own role in contributing to a better future?"

Reflections on Problems (5 minutes):

- "After the game, did you identify additional challenges or barriers your society faces in achieving sustainability and equity?"
- "Were there any moments during the game that stuck with you as particularly illustrative of real-world issues?"

Reflections on Policies (5 minutes):

- "Have you had any further thoughts about policies or actions that could help address the challenges discussed in the game?"
- "Did the game prompt you to think differently about specific policies, either positively or negatively?"
- **Closing**

Final Reflections (2-3 minutes):

- "Is there anything else you'd like to share about the game or its themes that we haven't discussed?"

Thank You and Next Steps (1 minute):

- Thank the participants for sharing their insights.
- Briefly outline how their feedback will be used to enrich the project and the broader goals of SPES.

Tips for Conducting the Interview

- **Tone**: Maintain a conversational and open-ended tone and listen well.
- **Probing**: Use follow-up prompts like "Could you elaborate on that?" or "What makes you feel that way?" to encourage deeper responses.

- **Timing**: Aim to keep the interview within 20-25 minutes.

5.2. Documentation of the individual interviews

The post-workshop individual interviews are recorded according to how they are done. If face-to-face with audio-recording equipment, if on the telephone or other means of teleconferencing, they can be recorded accordingly. These individual interviews are transcribed in the local language. Summaries can be made in English.

Section 2. After the workshop: Reporting

This second section of the manual addresses how the data produced in the workshop are handled, analysed and written up to a national report.

6. Writing the national report

The end results of the workshops are national reports from the national workshops. These reports shall inform deliverable 8.4 – a working paper with the tentative title: "The transition from below: Collectively agreed perspectives on aspirations, problems, and policy solutions".

6.1. Template for the national workshop reports

The national reports in English should not be more than 20 pages (1,5 line spacing, 12-point font size) contain the following:

Introduction:

- About the SPES project in the national context,
- the aim of the national reports,
- the research question that the report should inform.

Background:

- The national context relevant for SPES and the research question (parameters from international indexes concerning economy, social and environment, including vulnerability to climate change). This is filled in by the OsloMEt and UniFi using the indicators framework developed in SPES.
- Short historical recap of climate policies evolution in the country with focus on particular challenges (cf. SPES report 2.2.)

Method: description of how the workshops were implemented:

  - game played,
  - focused group discussion,
  - post workshop interviews,
  - ethical reflections.

Thematic analysis:

Thematic analysis is used to systematically interpret the rich, qualitative data generated during the workshops. Given the participatory nature of the LGJ game sessions and focus group discussions, participants express a wide range of perspectives on sustainability transitions, shaped by their lived experiences. Thematic analysis allows us to identify patterns and recurring themes across different national contexts while preserving the complexity and depth of participant contributions.

Our approach involves several key steps. First, we code the raw data—field notes, transcripts, and participant-generated outputs—to identify key topics of discussion. These topics are then grouped into broader themes that reflect central concerns related to sustainability transitions, such as energy justice, green employment, and social inclusion. Each theme is analysed using the SPES framework, organizing insights into aspirations, problems, and policy solutions to ensure a structured and policy-relevant interpretation. By using this method, we can compare findings across workshops, highlighting both national specificities and common challenges that inform Deliverable 8.4. Thematic analysis thus serves as a bridge between on-the-ground participant perspectives and broader policy discussions, ensuring that the voices of socio-economically disadvantaged groups are meaningfully incorporated into sustainability debates.

Theme	**Aspirations** ***(Desirable Futures & Collective Goals)***	**Problems** ***(Barriers & Challenges in the Transition)***	**Policy Solutions** ***(Opportunities & Potential Pathways for Change)***
(Example - to be replaced) **Energy Justice**	Affordable & accessible renewable energy. Community-led energy projects.	Energy poverty & high costs. Unequal access to green energy infrastructure.	Targeted subsidies for vulnerable households. Investment in decentralized renewables. Citizen participation in energy governance.
Green Employment & Just Transition	Secure & well-paid green jobs. Opportunities for re-skilling & up-skilling.	Job insecurity in transitioning industries. Lack of training programs for fossil fuel workers.	Government-funded retraining programs. Incentives for green business development.
…			
…			
…			

Concluding Reflections:

Appendix: pictures illustrating particularly interesting outcomes which emerged on the board during the LGJ game sessions.

Section 3.

7. Appendices

The more thorough and comprehensive description of the event cards, dice and wheel are included here because they take up a lot of space in the document, which disturbs the flow.

7.1. Events

The event cards come into play at the end of each round. Using the wheel, the GM ends up applying either a collective event card (80% chance) that affects all players or an individual event card (20%) that only affects that player.

If the collective events are printed/Written up on cards, the player draws from the deck of cards. If the events are only in the list, the GM uses a dice to select the event.

Concerning the individual events, the GM chooses a fitting card based upon the role and situation of the concrete individual players.

The consequences of the events for the player(s) are "modelled" by the players and the GM through discussion. The GM can do this by asking the player(s) what they think is the impact of the event on them. The "suggested impact" category in the table below is there to give the GM some suggestions as to what the local and societal impacts of the event in question might be. However, the players should be encouraged to discuss what *they* see as the impact, considering their own roles and perspectives.

7.1.1. Collective event cards (LGJ)

These event cards highlight real-world trade-offs and amplify the tension between individual and collective interests, prompting participants to reflect on sustainability challenges and decision-making complexities. The consequences in the game for the collective and the individual players are discussed by the players and translated into whites and greys.

Some spots are left open for the national teams to make event cards that are adapted to the local context.

Category	Title	Description	Suggested impact
Policy Events	Carbon Tax Reform	The government introduces a carbon tax to fund renewable energy projects.	Businesses and individuals face higher costs, but long-term environmental benefits are expected.

Category	Title	Description	Suggested impact
	Urban Expansion Policy	Zoning laws are relaxed, enabling urban development.	Economic growth is stimulated, but environmental and social strains increase.
	Rural Infrastructure Act	A policy funds new roads and utilities in rural areas.	Enhances rural connectivity and opportunities but comes at the expense of biodiversity loss, disrupting local ecosystems.
	Local green energy investments	Local government requires green energy investments.	Society sees a shift toward sustainability, with short-term sacrifices for long-term gains. Local pot can be used only for green loans
	Tax on above-average energy consumption	Higher value-added tax on above-average private energy consumption. (A policy that tries to model a policy that adopts the polluter-pays principle).	Incentive to save energy and let the well-off pay for the transition.
	Country specific policy event card	Insert description of policy event	Think about the impact it may have (aid for GM)
	Country specific policy event card	Insert description of policy event	Think about the impact it may have (aid for GM)
Climate Events	Severe Drought	A prolonged drought reduces water supply, affecting agriculture and industry.	Communities struggle with resource scarcity, leading to economic and social challenges. Assembly has to choose a local policy to mitigate the effects.
	Flooding in Urban Areas	Heavy rains flood the city core, damaging infrastructure.	Public services are disrupted, requiring community-wide recovery efforts.
	Wildfire Outbreak	Wildfires spread through the wilderness, threatening nearby areas.	Loss of biodiversity and property, with increased pressure on emergency services. See modeling of fires on GM manual.
	Invasive Species Spread	A non-native species disrupts local ecosystems.	Agricultural productivity and native species are threatened, sparking ecological concerns.

Category	Title	Description	Suggested impact
	Energy Blackout	A storm damages the grid, leading to widespread outages.	Economic activities are halted, and social unrest may emerge as essential services are affected.
	Country specific climate event card	Insert description of climate event	Think about the impact it may have (aid for GM)
	Country specific climate event card	Insert description of climate event	Think about the impact it may have (aid for GM)

How to Use These General Descriptions

- **Game Master's Role**:
  - The GM can use the Impact descriptions as prompts to guide discussions about societal responses, trade-offs, and consequences.
  - Encourage players to suggest solutions or explore the ripple effects of the events.
- **Player Engagement**:
  - Players can debate or collaborate on responses to the events, using their roles to frame their contributions.

7.1.2. Individual cards

If the wheel of fortune assigns an individual card, the GM chooses a card and reads it to the player, who is then asked to reflect upon what impact this has for her/him. Subsequently, the GM translates how this event affects the player's situation in white and grey currencies.

Title	Description	Suggested Impact
Unexpected Wealth	You inherit a small fortune from a distant relative.	Encourages investments or innovation but raises questions about fairness and equity.

Technological Breakthrough	Choose one technological breakthrough that you think would help the transition to a sustainable society.	Depending on the players' suggestion, think collectively about potential ecological, economic and/or social burdens.
Community Dispute	A conflict arises between your community and a neighboring group over land use. How do you react? Do you join the conflict?	Challenges collaboration and resource-sharing while highlighting the need for mediation skills.
Environmental Concern	You discover that your business operations are causing local pollution.	Sparks a dilemma between profit and environmental responsibility.
Health Crisis	A personal health emergency affects your productivity and resources.	Illustrates vulnerabilities in individual resilience and systemic health infrastructure.
Volunteer Opportunity	You are invited to join a local sustainability initiative. What do you choose?	Encourages civic engagement and collaboration, with potential long-term benefits.
Market Crash	An economic downturn disrupts your financial plans. How are you affected? What do you do?	Forces reevaluation of strategies and highlights systemic economic dependencies.
Unexpected Criticism	Your public actions are criticized for being unsustainable.	Encourages self-reflection and adaptation to align better with societal values.
Innovative Partnership	A neighboring group proposes a partnership on a sustainability project. How do you contribute?	Offers opportunities for resource-sharing and innovation but requires negotiation skills.

How to Use the Table in Gameplay

- **Game Master's Role**:
    - Use these events to challenge individual players, tailoring outcomes to the narrative.
    - Facilitate discussions about the broader implications of the events.
- **Player Engagement**:
    - Reflect on the event's impact and propose strategies to address challenges or leverage opportunities.

7.2. How to use the dice

The dice are a critical tool in *LGJ*, introducing chance, variability, and decision-making opportunities. This guide explains how dice are used across different aspects of the game.

Types of Dice and Their Purpose

The game uses a variety of dice to simulate diverse outcomes and probabilities:

- **d4, d6, d8, d10, d12, and d20**: Represent varying levels of randomness and complexity.



- **Standard Nomenclature**:
    - `1d6`: Roll one six-sided die.
    - `2d10`: Roll two ten-sided dice and sum the results.

These dice can be found in shops. Alternatively, dice apps can be used. See options here

Core Uses of Dice in the Game

1. **Resource Allocation**:
    - At the start of the game, dice are rolled to determine initial resources:
        - **2d6 for whites** (positive tokens).
        - **1d6 for greys** (negative externalities).
2. **Land Distribution**:

    - Players roll **1d4 + 1** to calculate the number of land units they own at the start.
3. **Activity Success Rates**:

    - Dice rolls determine the outcomes of player actions, such as investments or inventions.
    - For example, when setting up a new venture, the GM might ask for a **1d10 roll**, with results interpreted as follows:
        - **1-2**: Unsuccessful start, requiring additional resources or effort.
        - **3-4**: Basic success with modest yield (e.g., `1d4 whites`).
        - **5-6**: Balanced success with potential for improvement (e.g., `1d4 + 1 whites`).
        - **7-8**: Strong success with variability (e.g., `1d6 + 1 whites`).
        - **9**: Major success (e.g., `1d8 + 2 whites`).
        - **10**: Exceptional success, potentially leading to bonuses.

Invention Outcomes:

- Inventions use a **d20** so you can model “breakthrough” giving a very small chance (es 5%).
- The GM defines a success range (e.g., rolls of **18-20** result in success) and incorporates narrative factors like cost and societal impact.

Event Outcomes:

- When an event occurs (e.g., collective or individual), players roll dice to determine its specific impact, based on preset tables or GM discretion.

Guidelines for Game Masters (GM)

1. **Interpretation of Rolls**:

    - Use the results as prompts for storytelling and collaborative decision-making.
    - Modify outcomes based on contextual factors or player input to enhance immersion.
2. **Flexibility and Chaos**:

    - The GM may adjust dice requirements (e.g., number or type) to suit evolving scenarios.
    - Use dice creatively to simulate unexpected developments or resolve disputes quickly.
3. **Encouraging Player Engagement**:

    - Explain how dice results influence the game to ensure transparency.
    - Allow players to suggest interpretations or implications of their rolls.

Practical Tips for Players and GM

1. **Use Physical or Digital Dice**:
    - Physical dice enhance tactile engagement.
    - Apps or online tools (e.g., Mighty Dice) can handle complex rolls like `60d6 + 3d4 - 1d100`.
2. **Document Rolls**:
    - Keep track of significant rolls to maintain narrative continuity.
3. **Balance Time**:
    - Limit time spent on dice rolls to maintain gameplay momentum. Consider using a timer if necessary.

7.2. How to use the wheel

This tool allows your master to map distribution laws of events or decide about directions, for example, of winds. We made one that can be written upon with whiteboard markers to use it flexibly. It is usually specified in the scenarios how to set it up, or it is left to the master. The pivot can be made with a pinboard pin, and to operate it, keep it in the palm of your hand and tick the moving arrow.

7.3. Information letter to participants

This is a draft proposal and needs to be adapted to the national context.

SPES | Sustainability performances, evidence & scenarios

How can productivity, equity, environmental sustainability, participation and empowerment, and human security come together to drive the transition towards sustainable human development in Europe?

This is an invitation to contribute to a sub-study in the research project "SPES - Sustainability Performance, Evidence and Scenarios". As a participant in this study, you will take part in an interactive game and a focus group exploring sustainable development. The activity lasts approximately 4 hours and you will have the opportunity to contribute with your thoughts and experiences on the topic. Participation is voluntary and your personal data will be treated confidentially in accordance with applicable regulations.

This information letter explains the purpose of the study and what participation means for you. Feel free to contact us if you have any questions!

Researcher at University, Name: +47 number/ name@institution.no

Purpose of the sub-study

This study is part of the SPES project, which explores how sustainable human development can be promoted by balancing productivity, equity, environmental sustainability, participation and security. The focus of this sub-study is to understand the experiences and perspectives of civil society organisations and individuals with different economic and social interests and backgrounds.

We invite you to participate in a group interview where we discuss these topics. As part of the group interview, we will play a game that simulates the challenges of the transition to a sustainable low-carbon society. The group interview will be an arena for sharing experiences, perspectives and ideas that can help develop more accurate and inclusive solutions. Your insights are valuable in shedding light on the challenges and opportunities related to these issues. The results will be used in scientific social analyses and will be disseminated to interested social actors in Norway and at European level. Similar group interviews will be conducted in three other European countries (Italy, the Netherlands and Hungary) and three countries in the Global South (Kenya, Pakistan and Sri Lanka).

Who is responsible for the research project?

OsloMet - Metropolitan University is responsible for the Norwegian part of the project.

Why are you asked to participate?

We want to include the insights and experiences of people and groups with different vulnerabilities in the transitions society is facing. Through the group interviews, we hope to gain a better understanding of what is perceived as important goals and effective policy instruments to support the transition towards more environmentally sustainable societies.

Your participation will provide us with valuable insights into how governments can better balance climate action with economic and social considerations, from different perspectives and population groups. The information gathered will help develop policy recommendations that are more targeted to the needs and challenges faced by different groups.

What does it mean for you to participate?

You will participate in an activity consisting of a game and a discussion. The activity lasts approximately 4 hours in total and includes:

1. **Play the board game "Le Grand Jeu"** (approx. 2.5 hours): An interactive role-playing game that invites participants to explore different policy challenges regarding the transition to low-carbon societies and the relationship between economic, social and ecological sustainability
2. **Group interview** (approx. 1 hour): A conversation where participants share reflections and experiences from the game, and discuss the themes of the study in light of their own perspectives.

In addition, you can choose to participate in:

3. **Follow-up interview (optional)** (max 30 min): All participants are given the opportunity for a follow-up interview where they can share their experiences and reflections on the participation itself and the topics discussed. The follow-up interview is an optional part of the study and does not affect your participation in the game or the group interview.

The event will take place at OsloMet - Metropolitan University's premises (Pilestredet campus).

We will be filming and audio recording throughout the activity to document the conversations and game play. The film material will primarily be used to ensure that we capture details from the discussions, while the audio recordings will be used as a basis for analyses and reporting.

We will take some pictures during the gaming activities but we will avoid faces so that no person will be recognisable.

You don't need to prepare, and you choose how much you want to say during the discussion.

We really appreciate your time and effort!

Participation is voluntary

Participation in the project is voluntary. If you choose to participate, you can withdraw at any time without giving any reason and all your personal data will then be deleted. There will be no negative consequences for you if you do not want to participate, or later choose to withdraw.

Since your data is part of a larger context together with other participants and it is difficult to separate it out, it is important for us that you are well informed about what you are consenting to by participating.

Your privacy - How we store and use your personal data

Purpose of the programme

Your personal data will only be used for the purposes described in this information letter.

How the data is stored

We will utilise TSD (Services for Sensitive Data) for the secure storage of all material containing personal data. This applies to

- Audio and video films: Audio recordings (from game, group interview and follow-up interview) and video recordings (from game and group interview)
- Transcriptions: The audio recordings are transcribed to text but all directly and indirectly identifying personal data is removed before they are taken out of the TSD secure zone.
- Video films: Used to supplement observation notes and will be deleted as soon as transcripts are available.

Anonymisation and sharing

- We remove all personal data from transcripts and summaries of interviews with the exception of job title/employer information. However, it is important to be aware that there may be a risk of recognition by virtue of your statements if you possess knowledge or opinions that few people possess.
- At the end of the project, anonymised data can be stored in an approved research data archive for future use in other research environments.

What happens to the data when the project ends?

- All personal data (audio, video, name and contact information, etc.) will be deleted as soon as possible, at the latest by 30 September 2025.
- If you withdraw your consent, we will delete all personal data relating to you, but anonymised data from group activities may still be used.

Your rights

As long as you can be identified in the data material, you have the right to:

- access to what information we process about you
- to have personal data about you deleted
- to have information about you that is incorrect or misleading corrected
- to receive a copy of the information, and
- complain to the Norwegian Data Protection Authority about the processing of your personal data

What gives us the right to process personal data about you?

- We process information about you based on your consent.
- On behalf of OsloMet - Metropolitan University, SIKT Knowledge Sector's service provider has assessed that the processing of personal data in this project complies with data protection regulations.

Where can I find out more?

If you have any questions about the study, or wish to exercise your rights, please contact us:

- Researcher Name, University: name@institution.no
- Data Protection Officer at University: name/surname,
- SIKT - The Knowledge Sector's Service Provider by email name@sikt.no
- or phone:
- Sincerely yours,
- Researcher : Name
- Project manager

Consent to participate in the SPES project

- I have received and understood the information about SPES and have been given the opportunity to ask questions. I agree to:
- to participate in the working seminar (game session followed by a group interview) on sustainable transition, which will be video and audio recorded
- to participate in an individual interview 3-5 days after the workshop, with a recording of the conversation
- that personal data about me will be stored and processed in accordance with OsloMet's privacy policy until [insert date].

I understand and agree to

that my opinions (including political views and any union affiliation) and insights may be reproduced in the written material (transcripts), even though they may ultimately be linked to my person and lead to recognition if I possess knowledge or opinions that few other people have.

(Signed by project participant, date)

7.4. Vocabulary

General Concepts

- **The research manual**: This document that
- **The workshop**: the overall research activity described in this manual consisting of playing LGJ; post game group interviews/focused discussions and individual follow up interviews.
- **Serious game**: A serious game can be understood as a game that is played with instrumental value such as learning in addition to the enjoyment of playing in itself. Le Grand Jeu with the specific scenario studied for the SPES project as described in this manual
- **Le Grand Jeu (LGJ)**: A participatory, narrative-driven serious game designed to simulate societal dynamics and encourage discussion on sustainability, equity, and innovation within the context of planetary boundaries.
- **The group interview/focus-group discussion**:
- **The individual follow up interview**:
- **SPES Scenario**: A tailored version of LGJ for the SPES project, focusing on balancing productivity, equality, sustainability, and participation in a fictional society.

Concept related to Le Grand Jeu

Roles

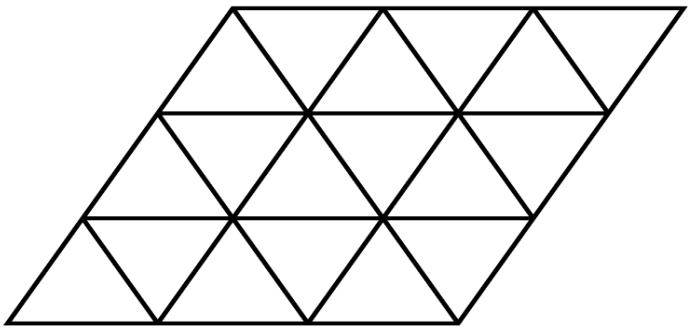
- **Game Master (GM)**: The facilitator who guides the game, ensures rules are followed, manages events, and supports player creativity.
- **Players**: Participants who assume individual or group roles, making decisions and navigating the challenges presented in the game environment.
- **Observer:** the researcher whose task is to assist the game master observing the facts and events to inform the discussion after the game and the report of the game.

Game Environment

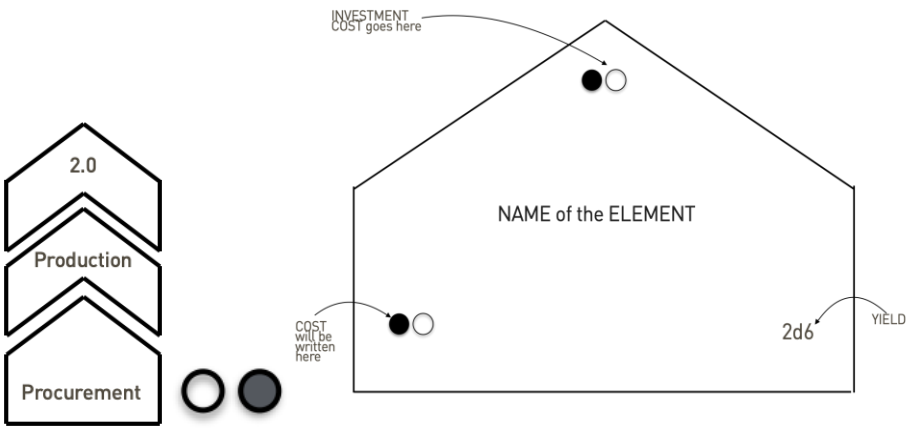
- **Urban Core**: A dynamic, resource-rich area offering services like schools, hospitals, and financial institutions but facing issues such as inequality.
- **Rural Area**: A greener, less serviced zone characterized by a sense of community and lower living standards.
- **Wilderness**: Untamed areas without infrastructure, where players must independently build and sustain their activities.
- **On the Grid**: Areas connected to basic infrastructure like water, electricity, and the internet.
- **Off the Grid**: Isolated locations lacking public infrastructure, requiring players to self-supply essential services.

Game Elements

- **Whites**: Tokens representing positive outcomes such as wealth, resources, energy, or knowledge.
- **Greys**: Tokens symbolizing negative externalities like pollution, resource depletion, and social costs.
- **Social Tokens (Reds)**: Represent social value created or destroyed within the society, assessed collectively by players.
- **The Grid**: A network overlay on the game board that determines access to infrastructure.



- **The Elements**: You find in the box two types of pieces that are to be used to design the elements of your activity. Elements are to be organised on 3 levels: procurement, production, 2.0. Elements can be combined in lines of production, individually or cooperatively or under any other form of agreement you can come up with. The elements are to be written upon with a white board marker, you will find some in the box



Game Mechanics

- **Game:** The complete session of Le Grand Jeu, encompassing multiple rounds, with the goal of collaboratively building a sustainable society while navigating individual and collective challenges. A game typically lasts 4–5 rounds or until specific conditions (e.g., ecological collapse) are met.
- **Round:** A full cycle of gameplay during which each player takes their turn, events are resolved, and the societal state is assessed. Rounds simulate the progression of time and collective decision-making**.**
- **Turn:** The time within a round when a single player acts. During their turn, the player can choose one action (e.g., invest, invent, call an assembly) and resolve its outcomes.
- **Invest**: An action where players expand operations or improve existing activities by allocating resources.
- **Invent**: A creative action where players propose new solutions to societal challenges, evaluated for feasibility.
- **Call an Assembly**: A collective decision-making process where players propose and vote on policies or actions.

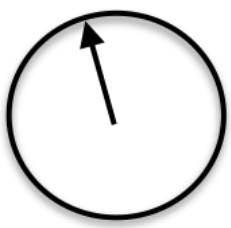
Game Tools

Event Cards: Predefined scenarios, either individual or collective, that introduce challenges or opportunities to the game.

Dice (d4, d6, d10, etc.): Used to introduce randomness and determine outcomes of player actions or events.d4 refers to dice with four sides; d6 refers to dice with 6 sides etcs



- **The Wheel**: A spinning tool to trigger events or distribute random effects, adding unpredictability to the game.



- **Game Board**: The physical representation of the SPES society, divided into urban, rural, and wilderness areas.

Key Concepts

- **Ecological Ceiling**: A limit (50 greys) representing the maximum tolerable negative externalities before societal collapse.
- **Social Value Scale**: The self-evaluated value of solidarity and community within the group playing at a scale between -10 and 10
- **Sustainability Transition**: The shift towards systems that balance economic productivity, social equity, and environmental preservation.

8. References

Bonelli, F. (2024) Melting Down Identity: Towards a Pedagogy of Complexity, June 2024, in Achour, M, Manai, A., Ed., The Encounter Between Identity and Innovation: An explosive brake of creativity, Edition D.A.C, Berlin 2024.

Gugerell, K. (2023). Serious games for sustainability transformations: Participatory research methods for sustainability-toolkit# 7. *GAIA-Ecological Perspectives for Science and Society*, *32*(3), 292–295.

Kriz, W. C. (2010). A systemic-constructivist approach to the facilitation and debriefing of simulations and games. *Simulation & Gaming*, *41*(5), 663–680.

Petz, M., Bonelli, F., & Eskelinen, T. (2024). Le Grand Jeu and the potential of money games for exploring economic possibilities. *Degrowth Journal*, *2*. https://doi.org/10.36399/Degrowth.002.01.01

Salliou, N., Bruley, E., Moreau, C., Luthe, T., Blanco, V., Lavorel, S., & Grêt-Regamey, A. (2021). Game of Cruxes: Co-designing a game for scientists and stakeholders for identifying joint problems. *Sustainability Science*, *16*(5), 1563–1578.

Speelman, E. N., Escano, E., Marcos, D., & Becu, N. (2023). Serious games and citizen science; from parallel pathways to greater synergies. *Current Opinion in Environmental Sustainability*, *64*, 101320. https://doi.org/10.1016/j.cosust.2023.101320

SPES

Sustainability
performances,
evidence & scenarios

www.sustainabilityperformances.eu